Firewall Basics Theory

Task 1: Outbound Rule to Block Spotify Desktop App

Step-by-Step Instructions:

1. **Open Advanced Firewall Settings:**
    - Press Windows Key + R, type **wf.msc**, and press **Enter** to launch **Windows Defender Firewall with Advanced Security**.
2. **Start the Outbound Rule Wizard:**
    - In the left panel, click on **Outbound Rules**.
    - In the right **Actions** pane, click on **New Rule...**.
3. **Select Rule Type:**
    - Select **Program** and click **Next**.
4. **Specify the Spotify Executable Path:**
    - Select **This program path**, click **Browse...**, and navigate to where Spotify is installed:
        - Standard path: %AppData%\Spotify\Spotify.exe (or C:\Users\<YourUsername>\AppData\Roaming\Spotify\Spotify.exe).
    - Click **Next**.
5. **Set Action:**
    - Select **Block the connection** and click **Next**.
6. **Assign Network Profiles:**
    - Check all three profiles: **Domain**, **Private**, and **Public**. Click **Next**.
7. **Name the Rule:**
    - Name: Block Spotify Outbound Traffic.
    - Description: Blocks all outgoing internet requests from the Spotify desktop app.
    - Click **Finish**.

- **Verification:** Open Spotify and attempt to play a song or load playlists. The application will immediately show an offline error (e.g., *"No internet connection / Error Code: 4"*).

Task 2: Inbound Rule to Allow Google Chrome on Port 443 (HTTPS)

Step-by-Step Instructions:

1. In the **Windows Defender Firewall with Advanced Security** (wf.msc) window, click **Inbound Rules** in the left sidebar.
2. In the right **Actions** pane, click **New Rule...**.
3. **Rule Type:** Select **Custom** and click **Next**.
4. **Program:**
    - Select **This program path**, click **Browse...**, and select Chrome:

C:\Program Files\Google\Chrome\Application\chrome.exe

(or C:\Program Files (x86)\Google\Chrome\Application\chrome.exe).

    - Click **Next**.
5. **Protocol and Ports:**
    - **Protocol type:** Choose **TCP**.
    - **Local port:** Choose **Specific Ports** and enter **443**.
    - **Remote port:** Leave as **All Ports**.
    - Click **Next**.
6. **Scope:**
    - Leave both local and remote addresses set to **Any IP address** and click **Next**.
7. **Action:**
    - Select **Allow the connection** and click **Next**.
8. **Profile:**
    - Ensure **Domain**, **Private**, and **Public** are checked. Click **Next**.
9. **Name & Finish:**
    - Name: Allow Chrome Inbound Port 443 HTTPS.
    - Click **Finish**.

Task 3: Step-by-Step Guide: Blocking All Traffic from a Specific IP (123.45.67.89)

Plaintext

[Windows Defender Firewall (wf.msc)]

└── Inbound Rules (or Outbound Rules)

└── New Rule...

├── 1. Rule Type: Custom

├── 2. Program: All programs

├── 3. Protocol & Ports: Any

├── 4. Scope: Remote IP -> 123.45.67.89 <-- (KEY STEP)

├── 5. Action: Block the connection

└── 6. Name: Block Malicious IP - 123.45.67.89

Detailed Execution Steps:

1. **Open Advanced Firewall Console:** Press Win + R, type wf.msc, and press **Enter**.
2. **Select Rule Direction:**
    - Right-click **Inbound Rules** (to prevent incoming traffic from that IP) or **Outbound Rules** (to stop your PC from reaching that IP), then click **New Rule...**.
3. **Choose Custom Rule:** On the *Rule Type* page, click **Custom** and select **Next**.
4. **Apply to All Programs:** On the *Program* page, choose **All programs** and select **Next**.
5. **Protocol & Ports:** Leave *Protocol type* set to **Any** and select **Next**.
6. **Configure the Scope Tab (Crucial Step):**
    - Under **Which local IP addresses does this rule apply to?**, leave **Any IP address**.
    - Under **Which remote IP addresses does this rule apply to?**, choose **These IP addresses**.
    - Click the **Add...** button.
    - In the popup box, select **This IP address or subnet**, type **123.45.67.89**, and click **OK**.
    - Click **Next**.
7. **Specify Action:** Select **Block the connection** and select **Next**.
8. **Select Profiles:** Keep **Domain**, **Private**, and **Public** checked. Click **Next**.
9. **Name and Save:**

- Name: Block Malicious Host - 123.45.67.89.
- Description: Drops all network packets originating from or destined to remote IP 123.45.67.89.
- Click **Finish**.

Task 4: Hardware Firewalls vs. Software Firewalls

